

REALM / RAG / RAP — Derivation Notes

Deriving REALM's gradient mechanism from scratch, comparing it against RAG, then evaluating a cross-modal RAP research proposal. Organized as: my derivation path → where I got stuck → the correct mechanism → experimental evidence → proposal assessment.

1. Starting point: what REALM is

REALM (Guu et al., 2020, ICML, Google) is the founding work of the "retrieval-augmented" line.

Core claim: don't cram all world knowledge into model parameters (that forces ever-larger models, and you can't tell what knowledge lives where). Instead, let the model **explicitly retrieve** from an external text corpus (e.g. all of Wikipedia) and use the retrieved documents to help it predict.

Three selling points: interpretable (you see which document was used), updatable (swap the corpus without retraining), and the corpus can be far larger than what parameters could hold.

The signature contribution: for the first time, train the retriever during pre-training end-to-end, with gradients backpropagating through the retrieval step. The learning signal is masked language modeling (fill the masked word).

2. My derivation, and where I got stuck (this part is the point)

Opening (right direction)

- reader input is $[z_i ; x]$ (retrieved doc z_i concatenated with input x), d_{model} wide
- L is the cost function, $\partial L / \partial z_{ij}$ is known
- goal: pass this gradient to the retriever

First wall: $\partial L / \partial z$ cannot reach the retriever

Realized on my own: - z_{ij} is the document token embedding on the **reader (BERT) side** - the retriever's two-tower uses **$g(z)$** — the vector output by the doc encoder - **these are not the same vector**; there is no shared computation graph between them → so $\partial L / \partial z_{ij}$ cannot flow to $g(z)$. This path is dead. (correct call)

Second wall: differentiating w.r.t. z is itself the wrong move

Briefly tried: $z' = z - lr \cdot \partial L / \partial z$ (update the document vector), then $\text{sigmoid}(g(z) \cdot f(x))$ as an independent score. Noticed "this is independent, it can't use the llm's result" — correct instinct, but the reason for being stuck is **still staring at differentiating w.r.t. z** . The retriever's parameters are the weights of f and g , not z , so you should not differentiate w.r.t. z at all.

The jump that unlocks it: differentiate w.r.t. the retrieval probability

What connects llm and retriever isn't the gradient on z — it's the equation **written much earlier when deriving the reader part**:

$$p(y|x) = \sum_i p(y|x, z_i) \cdot p(z_i|x)$$

- $p(y|x, z_i)$: probability the reader fills the mask correctly given doc z_i (the llm's part)
- $p(z_i|x)$: the retrieval weight of doc i (the retriever's part) = softmax over top-k ($f(x) \cdot g(z_i)$)
- the two are **multiplied inside the same Σ** → when you differentiate w.r.t. $p(z_i|x)$, the llm's result comes in automatically

→ the gradient does not pass through the document embedding; it passes through the **retrieval weight** into the two-tower. This is the heart of REALM.

Three engineering difficulties, all pointed out during the derivation:

1. **Retrieval is discrete** (choose from millions) → top-k approximation: keep only top-k, treat out-of-set $p \approx 0$
2. **Full-corpus softmax is too expensive** → normalize only within top-k (the jump of "assume top-k is the entire volume")
3. **The index goes stale** (once the retriever updates, $g(z)$ changes) → asynchronous: rebuild the index only every few hundred steps (stale index)

Physical meaning (the closing step)

top-k are not fed to the reader with equal weight; each carries weight $p(z_i|x)$. The form of $\partial L / \partial p(z_i|x)$: **documents whose reader performance is above average get their retrieval weight upweighted**; worse ones downweighted. This is what the retriever implicitly learns — no explicit label; the supervision is hidden in "which document makes $p(y|x)$ higher".

3. My scheme vs REALM's scheme (a real alternative branch that split off mid-derivation)

Mid-derivation, an alternative **of my own** appeared, different from REALM, worth recording:

My route (two-stage, explicit pseudo-label): 1. update z to convergence via gradient → ideal document vector z' 2. find the nearest real document z to z' in the corpus 3. use z as a positive-sample label → contrastive learning to train the two-tower (query= x , positive= z^* , rest negative) 4. once the encoders are trained → all x benefit

This route works, and is a genuinely existing branch — close to DPR / distilling the retriever with downstream signal, not REALM. Effectively re-inventing hard-negative mining. - upside: uses "nearest-neighbor to grab one doc as label" to bypass full-corpus softmax; cheap - weakness: z' is "ideal", the corpus is "real"; when z' falls in a gap of the document distribution, the nearest neighbor z may not be relevant → *projection noise** (the projection error from continuous ideal to discrete real)

REALM's route (end-to-end, implicit supervision): - no label construction, no updating the

document z , no independent sigmoid scoring - softmax over top-k + differentiate the expectation Σ directly; one loss, one backward pass, trains reader and retriever simultaneously - clean, but pays the engineering cost of top-k approximation + async index

One-line comparison: | | my route | REALM | |---|---|---| | supervision | explicit pseudo-label (z^*) | implicit (hidden in Σ) | | gradient acts on | move doc z first, then nearest-neighbor | retrieval probability $\rightarrow$ two-tower params | | stages | two-stage (make label + train) | end-to-end, one loss | | discreteness | nearest-neighbor projection (noisy) | top-k truncation of full expectation | | cost | cheap | expensive |

Being able to see that "what I derived $\neq$ REALM" itself means both paths were touched, and the fork is understood: **it's about how to handle discreteness — projection vs full expectation.**

4. REALM vs RAG: mechanism and experiments

Mechanism differences

- **REALM:** train the retriever end-to-end during pre-training (MLM signal); the retriever is **frozen** during downstream fine-tuning
- **RAG (Lewis 2020):** use a pre-trained DPR retriever, do only partial end-to-end fine-tuning, reader is BART (generative); cheaper, open-source
- **Modern RAG (industry):** degenerated to — frozen embedding retrieval + frozen large LLM + concatenate into the prompt at inference. Simplest and cheapest, mathematically the "roughest"; popular because cheap / modular / updatable
- **RETRO (2022):** another branch — fuses retrieval via chunked cross-attention, retriever **frozen**. (Note: how the retrieved doc enters the model — REALM is "concatenate + probability marginalization", RETRO is "cross-attention"; I initially crossed the two up — REALM does not fuse via attention, it fuses via probability weighting)

Directly comparable experiments (2020, same period)

dataset	REALM	RAG
Natural Questions	40.4	44.5
WebQuestions	40.7	45.2
TriviaQA	~55	56.8

Conclusion: head-to-head at the same time, RAG is slightly ahead or on par; REALM did not cash in its theoretical advantage. The reason is not that the theory is wrong, but that this paper's implementation didn't let the advantage show: 1. RAG uses a stronger generator (BART); REALM is extractive with a weaker reader 2. REALM freezes the retriever during downstream fine-tuning, so the end-to-end advantage is not used 3. General domain (Wikipedia); a generic retriever is already good enough, so the marginal gain from joint training is small to begin with

The theoretical advantage is only cashed in by later work

- **Sachan et al. 2021** (fully end-to-end retriever training): answer extraction beats **both REALM and RAG by 3+ points**; top-20 retrieval accuracy beats DPR by 5 points → confirms "bringing the retriever into the objective and optimizing jointly is indeed better"
 - **RAG-end2end (Siriwardhana 2022, TACL)**: specifically tests domain shift (COVID/News/Conversations). Jointly training the retriever + updating the index **significantly** beats vanilla RAG; updating only the query encoder without updating the corpus encoding hurts → confirms "under cross-domain, the retriever badly needs to align, and joint-training value returns"
 - **RARE (2026)** breaks down RAG's gain over pure-parametric by domain: General-Wiki +7.1 / Finance +13.5 / Legal +19.7 / Patent +33.4 → **the more the domain is outside parametric knowledge, the larger the marginal gain of retrieval**
-

5. My judgment "REALM is theoretically superior", and its experimental confirmation

My judgment: treating retrieval as a variable brought into the objective and optimized jointly is, from a probabilistic view, closer to the true value and theoretically generalizes better; RAG is an **approximate substitute** that treats retrieval as a fixed external step.

The premise to attach (a correction): the size of "superior" depends on **how much the retriever needs to align for the downstream task/domain**. - General domain: a generic retriever is already good enough → marginal gain of joint optimization is small → eaten by engineering cost (hence RAG caught up in 2020) - Specialized / shifted domain: alignment need is large → joint-training advantage is significant (confirmed by Sachan / RAG-end2end / RARE)

Why the field retreated to the rougher RAG — two drivers (not just engineering): (a) engineering cost (full-corpus softmax + async index too heavy) (b) **parametric scaling worked too well** — large models proved that scaling knowledge into parameters works extremely well, weakening the necessity of "must train retrieval end-to-end" → direction of evolution: rougher but more practical. Mathematically beautiful ≠ should be used; when scaling reaches comparable results with a dumber method, the beautiful one is shelved.

6. RAP proposal assessment (doing cross-modal on top of REALM)

Proposal claim: pre-training data scale determines model capability, but many scientific domains (healthcare / life sciences / time series / climate / materials) lack "natural-language-scale" data. Use RAP (retrieval-augmented pre-training) to break the "capability $\propto$ data scale" dependency. Validation setting: structured EHR as the pre-training modality + biomedical text as the retrieval corpus.

The clever part of the framing

Re-interprets "RAP has been neglected" as: it was neglected because people only tried it where RAG also works (same modality, data-rich). The real gap is **data-scarce + cross-modal + unaligned** — there, RAG's three prerequisites (same modality, natural alignment, interchangeable as input) all fail, and only there does RAP have exclusive value.

The three-cell analysis I derived (drawing the applicability boundary of RAG/RAP)

setting	doable?	is RAP necessary?	proposal's opportunity
same modality (cross-domain)	just add data / RAG works	no	none
cross-modal aligned (image-text, CLIP exists)	multimodal RAG works	no (RAG suffices)	none
cross-modal unaligned (EHR→text, no pairs)	nobody has validated	yes (RAG can't)	yes, but no evidence

The core dilemma: where necessity exists (unaligned) is exactly where feasibility evidence is missing; where feasibility evidence exists (aligned) is exactly where necessity is missing. These two conditions have never been satisfied simultaneously in existing work.

Real problems in the proposal (reviewers will poke)

1. **The RETRO analogy is backwards:** the proposal cites RETRO's "saves an order of magnitude of data" as evidence, but RETRO **freezes the retriever** — it does not do the "jointly train embeddings" the proposal features. RETRO saves data precisely because it does NOT do the expensive thing.
2. **"Order-of-magnitude gain" lacks mechanism:** REALM over RAG in the original paper is 4-16% absolute accuracy, not an order of magnitude. Cross-domain may be larger, but "order of magnitude" needs a mechanistic argument, not a deduction from "theoretically superior".
3. **Compute cost is self-contradictory:** lists "high cost" as a reason RAP was neglected, then promises to "solve the compute challenge" with no how; and cross-modal joint training is only more expensive.
4. **The validation setting is soft:** EHR + text are semantically highly related — the setting most easily rebutted by "just distill it into structured knowledge and absorb into parameters". (But see below — from the alignment angle, this "relatedness" is actually an advantage.)

The key line of questioning (pushed step by step)

- **Q: If the retrieval corpus is large and training data is scarce, why not just put it into the training data instead of adding a retrieval layer?** A: Holds for same modality (just add it, RAP is redundant). Under cross-modal, "the retrieval corpus can't be used as training samples" is RAP's only defense — but the proposal never argues why "retrieve" beats "distill into parameters". And if the retrieval corpus is large enough to support order-of-magnitude gains, it IS "data" — the only difference is "different modality, can't be used directly as training samples". The proposal's viability rests **entirely** on this premise.
- **Q: A multimodal LLM eats image+text; retrieving an image and feeding it in is just cross-modal RAG, no?** A: Correct. Multimodal RAG can already retrieve cross-modal. So the proposal's prerequisites (a) same-modality and (c) interchangeable are falsified; only (b) natural alignment stands. RAG (even multimodal) assumes the retrieval source and query are in an

already-aligned space (CLIP); the proposal wants **unaligned** cross-modal — that is the only gap. Scope must narrow to "unaligned cross-modal".

warm-start: from "fatal flaw" to "a step with a plan" (the most important turn)

Initially I judged warm-start (cold-start alignment) to be the fatal flaw: unaligned + no paired data → the retriever at cold start doesn't know which text an EHR should retrieve → joint-optimization gradients are noise → may converge to a fake alignment.

But this judgment was too absolute, corrected by one rebuttal: alignment need not pre-exist; the training process itself can learn alignment, and the simplest is to embed a CLIP-style contrastive step. CLIP itself is the proof — image and text are not aligned at first; the alignment is trained via contrastive learning over 400M image-text pairs.

Corrected, precise judgment: - warm-start is not "can it be done", it's "**what signal to bootstrap alignment with**" - **EHR-text has weak paired signal**: each EHR record carries ICD codes, medications, lab values, mappable to medical ontologies (UMLS/SNOMED) and corresponding literature → alignment signal is nonzero - so "EHR+text are semantically highly related" (earlier a weakness) is **actually an advantage** — the paired signal exists, CLIP-style warm-start is feasible - the concrete plan should read: **first do weak supervision with structured codes + medical ontology, run CLIP-style contrastive pre-training to get an initial alignment, then enter RAP end-to-end joint training to refine**

This turn changes the proposal from "betting on an unvalidated hard problem" to "combining two mature techniques (CLIP alignment + REALM end-to-end) on structured medical data". The feasibility assessment is entirely different — the latter is assembling existing parts, far more feasible. The answer to "why has nobody done it" changes too: maybe not "can't be done", but "nobody connected these two lines on EHR". The novelty is in **the combination + application domain**, not in solving a dead problem from scratch.

Does multimodal have a direct conclusion

- Aligned multimodal (image-text): has a conclusion, doable — but that cell is served by RAG, RAP not needed
- Unaligned cross-modal RAP (EHR→text, end-to-end joint retriever training): **no direct conclusion**, the gap the proposal claims
- Note: you cannot use the experimental success of cross-domain (text→text) to back unaligned cross-modal — in the former, alignment is free / pre-exists; in the latter, alignment is the problem to solve. The difficulties are of different nature.
- "No conclusion ≠ opportunity": first rule out "nobody did it because it can't be done". With the CLIP-warm-start plan, it leans toward "genuine gap + feasible", but a thorough literature search is still needed to substantiate the novelty claim.

7. One-page conclusion

1. **REALM is mathematically more complete**: end-to-end, brings the retriever into the objective,

gradients flow via the retrieval probability (not through the doc embedding, not via attention). Theoretically closer to the truth and generalizes better — but the premise is "the retriever needs to align for the downstream".

2. **In 2020, head-to-head, RAG was slightly ahead** — because that REALM paper's implementation (frozen retriever downstream, weak reader, general domain) didn't cash in the theoretical advantage. The advantage was only confirmed by later fully-end-to-end (Sachan 2021) and domain-shift work (RAG-end2end / RARE).
3. **The field retreated to the rougher RAG**: engineering cost + parametric scaling working too well, two reasons jointly weakening the necessity of end-to-end.
4. **The RAP proposal**: the framing and gap are well-identified; the only cell with exclusive value is "unaligned cross-modal". The fatal-looking warm-start turns into a step-with-a-plan via "embed CLIP-style weak-supervision alignment to bootstrap" — the proposal essentially shifts from "solving a dead problem" to "combining mature parts (CLIP + REALM) on EHR". Still needs: narrow scope to unaligned cross-modal, write warm-start as the concrete CLIP plan, give "order of magnitude" a mechanism or downgrade it, do a thorough literature search to substantiate the gap.

(A methodological thread running through the derivation: separate "what is observed" from "what can be concluded" — high AUROC / theoretically-superior is only correlation or potential; cashing it in depends on implementation and experiments; continuous-ideal projected onto discrete-real always carries noise; necessity and feasibility must be argued separately, they cannot borrow from each other.)